VAR Model Summary — Part 1

Oct 10, 2026 · @Keehong Rhee

At a Glance

A VAR (vector autoregression) bundles several time series into one system so that each variable reacts to **the past of all variables**. The conclusions of Part 1 come down to three points.

1. **A VAR is an 'AR for vectors'.** Where a univariate AR looks only at its own past, a VAR also looks at the past of the other variables. Those links sit in the off-diagonal elements of the coefficient matrix (the cross effects).
2. **The constraints are stability and parsimony.** All eigenvalues of the coefficient matrix must lie inside the unit circle, and because the number of parameters grows with the square of the number of variables, the system must be kept small.
3. **A joint model opens up new questions.** A fitted VAR forecasts several variables at once, and we can then ask "who leads whom (Part 2, Granger causality)" and "how do shocks spread (Part 3, IRF and FEVD)".

1. Why Model Several Time Series Together?

Market variables influence one another, so modelling them one at a time throws away the information in those links. Interest rates move stock prices, oil prices move airline stocks, and a shock to one asset spills over into others.

What univariate models miss

The AR, ARMA and GARCH models we have studied so far all explain **a single series by its own past alone**. Even the GARCH of Week 5 modelled each asset's volatility separately. Yet portfolios, hedging and pairs trading all depend on "how several assets move together".

Modelling them jointly gives us three things.

- **Better forecasts**: the past of one variable helps forecast another (borrowing strength).
- **Lead–lag tests**: we can check statistically who moves whom first (Part 2).
- **A map of shock transmission**: we can trace how a shock to one variable spreads through the whole system (Part 3).

Everyday examples include short- and long-term interest rates, an individual stock and its sector index, and spot and futures prices (the starting point of pairs trading).

Episode: "Economists, stop pretending to know" — Christopher Sims (1980)

Macroeconomics in the 1970s relied on large structural models made up of hundreds of equations. These models rested on hundreds of untested theoretical restrictions (assumptions that certain

variables determine others). In his 1980 paper "Macroeconomics and Reality", Sims criticised these restrictions as "incredible" assumptions: theory had been dressed up as fact.

His alternative was simple.

- Treat all variables symmetrically. Theory does not designate any particular variable as a cause.
- Let each variable depend on the recent past of all variables.
- This minimal-assumption model is the VAR.

The VAR went on to become a standard tool of empirical macroeconomics and finance, and Sims received the Nobel Prize in Economics in 2011. The lesson is "model the relationships you can measure, not the ones you assume". We should, however, stay modest about whether what a VAR shows is 'causation'. Part 2 takes up this question.

2. The VAR(p) Model: An AR for Vectors

VAR(p) is AR(p) with a **vector** in place of the variable and **matrices** in place of the coefficients.

Starting point: AR(p) revisited

A univariate AR(p) explains today's value by its own p past values.

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

Definition of VAR(p)

Let Y_t be the vector of k variables stacked vertically. Then VAR(p) is as follows.

$$\mathbf{Y}_t = \mathbf{c} + A_1 \mathbf{Y}_{t-1} + A_2 \mathbf{Y}_{t-2} + \cdots + A_p \mathbf{Y}_{t-p} + \boldsymbol{\varepsilon}_t$$

Symbol	Size	Meaning
Y_t	k×1	Vector stacking the k variables at time t (e.g. a stock return, a sector return)
c	k×1	Constant term of each equation
A_i	k×k	Coefficient matrix for the i-th lag. Contains both own effects and cross effects
ε_t	k×1	Error (shock) vector. Mean 0, covariance matrix Σ
p	Integer	Lag order. Same meaning as the p in AR(p)

The key point is that **every variable is regressed on the past of every variable**. With one variable (k=1) the matrices become numbers, and VAR(p) reduces to AR(p). The VAR is therefore the natural generalisation of the AR.

Writing out a two-variable VAR(1)

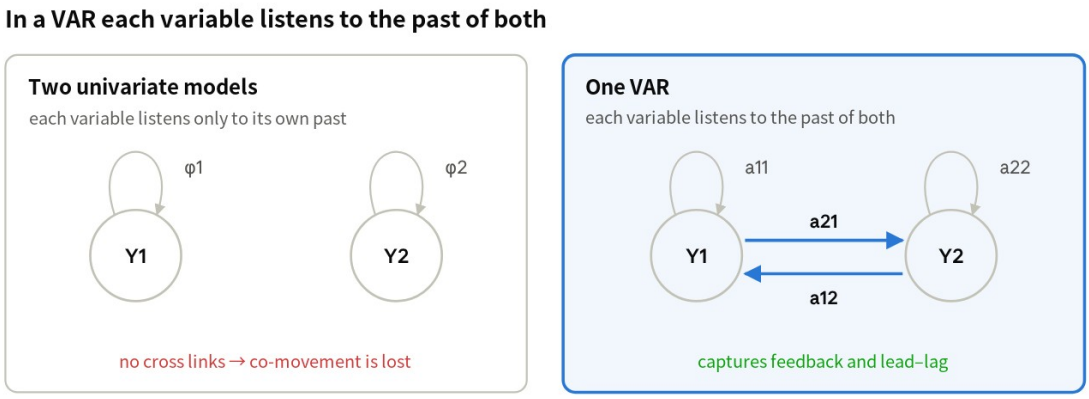
If matrix notation feels unfamiliar, write it out with two variables and one lag.

$$y_{1,t} = c_1 + a_{11}\, y_{1,t-1} + a_{12}\, y_{2,t-1} + \varepsilon_{1,t}$$
$$y_{2,t} = c_2 + a_{21}\, y_{1,t-1} + a_{22}\, y_{2,t-1} + \varepsilon_{2,t}$$

Collected into matrix form, the same equations read as follows.

$$\begin{pmatrix} y_{1,t} \\ y_{2,t} \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix}$$

- Each variable depends on **its own** previous value (a11, a22) and **the other variable's** previous value (a12, a21).
- If a12 ≠ 0, the past of Y2 helps forecast Y1. If a21 ≠ 0, the reverse holds.
- These cross terms are the whole reason for using a VAR. If all cross terms are 0, it is the same as running two unrelated ARs separately.



Two univariate models vs one VAR · own effects and cross effects

On the left are two ARs that look only at their own past (φ); on the right is a VAR with the cross arrows added (a12: Y2→Y1, a21: Y1→Y2). These two arrows are the information a VAR has beyond the univariate models.

3. How to Read the Coefficient Matrix

The coefficient matrix is a map that captures the link structure of the whole system on a single page. The diagonal elements are own effects; the off-diagonal elements are the interactions between variables.

$$A_1 = \begin{pmatrix} a_{11} & \overbrace{a_{12}}^{\text{cross}} \\ \underbrace{a_{21}}_{\text{cross}} & a_{22} \end{pmatrix}$$

Position	Element	Meaning
Diagonal	a11, a22	Own-lag effect. Plays the same role as a univariate AR coefficient
Off-diagonal	a12, a21	Cross effect. How much the past of one variable moves another

There are three things to look for.

- **Size**: large off-diagonal elements indicate a strong lead–lag relationship.
- **Sign**: positive (+) means the influence runs in the same direction, negative (−) in the opposite direction.
- **A diagonal matrix** means there is no interaction at all. The VAR is then the same as k unrelated AR models.

One caution. The coefficient matrix shows only **lagged** relationships. Movement together within the same period lives in the error covariance Σ, which is dealt with in Part 3 (orthogonalization of shocks).

4. The Reduced-Form VAR and Estimation

The VAR we have seen so far is a 'reduced form', and it can be estimated by OLS (ordinary least squares) equation by equation. It contains almost no theory, and estimation is surprisingly simple.

Reduced form: let the data speak

The reduced-form VAR makes only one assumption: "the variables depend on one another's recent past". This is the minimal-assumption model Sims had in mind.

- No variable is designated as the 'cause' of another.
- The equations only describe the correlation structure over time; they make no claim about economic mechanisms.
- It deliberately withholds judgement about the 'true' structure.

There is a price. We get dynamics and forecasts faithful to the data, but causal shocks such as an "interest-rate shock" cannot be read directly off the reduced form. That requires additional assumptions, which Part 3 covers.

Estimation: equation-by-equation OLS

Every equation in a VAR has **the same regressors** (the lagged values of all variables plus a constant). For example, both equations of a two-variable VAR(1) use the following regressors.

$$\mathbf{x}_t = (1,\ y_{1,t-1},\ y_{2,t-1})^\top$$

When the regressors are identical, estimating each equation by OLS gives the same result as estimating the whole system at once (a result from SUR theory). Equation-by-equation OLS is therefore an efficient estimator.

Estimation yields two things.

1. The coefficient matrices A_1, ..., A_p and the constant c
2. The residual covariance matrix Σ

$$\hat{\Sigma} = \frac{1}{T}\sum_{t=1}^{T} \hat{\boldsymbol{\varepsilon}}_t \hat{\boldsymbol{\varepsilon}}_t^\top = \begin{pmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{12} \\ \hat{\sigma}_{12} & \hat{\sigma}_2^2 \end{pmatrix}$$

The off-diagonal element σ12 of Σ measures how much the two variables' shocks arrive together **in the same period**. If it is not 0, the experiment "shock Y1 alone" becomes impossible, and this is the starting point of Part 3. In practice, software such as statsmodels fits the whole system in one go.

5. The Stability Condition and the Stationarity Prerequisite

A VAR is stable only when all eigenvalues of its companion matrix lie inside the unit circle (absolute value < 1). This is the multivariate version of the univariate unit-root condition |φ| < 1.

Intuition: shocks pass on by multiplying by A each period

In a VAR(1), the effect that a shock ε_t arriving today leaves on Y after h periods is A multiplied h times.

$$\frac{\partial \mathbf{Y}_{t+h}}{\partial \boldsymbol{\varepsilon}_t^\top} = A^h$$

When you keep multiplying by a single number, it shrinks to 0 if its absolute value is below 1 and explodes if it is above 1. For a matrix, the eigenvalue λ plays the role of that 'size'. The condition is therefore as follows.

$$A^h \to 0 \quad (h \to \infty) \iff |\lambda_i| < 1 \text{ for all } i, \qquad \det(A - \lambda I) = 0$$

For a 2×2 matrix, the eigenvalues follow directly from the trace and the determinant (det).

$$\lambda = \frac{\operatorname{tr}(A) \pm \sqrt{\operatorname{tr}(A)^2 - 4\det(A)}}{2}$$

For example, if A = [[0.5, 0.3], [0.1, 0.4]], then tr = 0.9, det = 0.17, and the eigenvalues are 0.63 and 0.27. Both are below 1, so the VAR is stable.

With several lags: the companion matrix

A VAR(p) is turned into VAR(1) form by stacking the lags, and the same condition is checked. For p = 2 it looks as follows.

$$\underbrace{\begin{pmatrix} \mathbf{Y}_t \\ \mathbf{Y}_{t-1} \end{pmatrix}}_{\boldsymbol{\xi}_t} = \underbrace{\begin{pmatrix} A_1 & A_2 \\ I & 0 \end{pmatrix}}_{F \text{ (companion)}} \begin{pmatrix} \mathbf{Y}_{t-1} \\ \mathbf{Y}_{t-2} \end{pmatrix} + \begin{pmatrix} \mathbf{c} + \boldsymbol{\varepsilon}_t \\ \mathbf{0} \end{pmatrix}$$

The lower block [I 0] is merely an identity that "shifts yesterday's values down by one slot". If all eigenvalues of this F lie inside the unit circle, the VAR(p) is stable.

Why it matters

	Stable VAR	Unstable VAR
Eigenvalues	All inside the unit circle	At least one on or outside the unit circle
Shocks	Die out over time	Are amplified
Path	Moves around a constant mean (stationary)	The whole system explodes (like a unit root)
Forecasts and IRF	Reliable	Unusable

So before looking at forecasts or IRFs, **check stability first**.

Stationarity comes first

As with ARMA, a standard VAR assumes stationary time series as input. The preparation goes as follows.

1. Run a unit-root test (Week 2, ADF etc.) on each series.
2. Difference trending series into returns or growth rates. For prices, use log returns.
3. Fit the VAR to the stationary series.

$$r_t = \ln P_t - \ln P_{t-1}$$ (the VAR takes returns r_t, not price levels P_t)

There is one exception. If two I(1) series are cointegrated, they are tied together in the long run, so differencing throws away the information in that long-run relationship. This case needs a separate vector error correction model (VECM), covered in Week 7.

6. Choosing the Lag p and the Curse of Dimensionality

Choose the lag p as the value that minimises an information criterion (AIC, BIC, HQ), but keep both p and k small, because the number of parameters grows with the square of the number of variables.

Information criteria: weighing fit against complexity

Adding lags always shrinks the residuals, but it also adds parameters to estimate and raises the risk of overfitting. An information criterion weighs "how small the residuals are" against "how many parameters were used" in a single number.

$$\mathrm{AIC}(p) = \ln|\hat{\Sigma}_p| + \frac{2}{T}\,pk^2$$

$$\mathrm{BIC}(p) = \ln|\hat{\Sigma}_p| + \frac{\ln T}{T}\,pk^2$$

$$\mathrm{HQ}(p) = \ln|\hat{\Sigma}_p| + \frac{2\ln\ln T}{T}\,pk^2$$

- The first term $\ln|\hat{\Sigma}_p|$ is the (log) determinant of the residual covariance matrix; the better the fit, the smaller it is.
- The second term is the penalty. Adding one lag adds k^2 coefficients.
- The **smaller** the value, the better. Fit the model for several p and pick the p with the minimum.
- With 8 or more observations, $\ln T > 2$, so the BIC penalty is heavier than the AIC penalty. BIC therefore tends to pick shorter lags. When the two disagree, BIC's simpler model is often steadier in out-of-sample forecasting.
- After choosing, check that no autocorrelation is left in the residuals. If some remains, the lag is too short.

The curse of dimensionality: parameters grow as k^2

With k constants plus one k×k matrix per lag, the number of parameters to estimate is as follows.

$$\#\text{params} = \underbrace{k}_{\text{constants}} + \underbrace{p\,k^2}_{\text{coefficient matrices}}$$

Number of variables k	p = 1	p = 2	p = 4
2	6	10	18
3	12	21	39
5	30	55	105
10	110	210	410

Doubling the number of variables roughly quadruples the coefficients. With 10 variables and 4 lags there are 410 to estimate, and with 10 years of monthly data (T = 120) that means fitting 41 coefficients per equation with 120 observations. A VAR is therefore kept **small and of low order**. Parsimony matters far more than in the univariate case.

7. Forecasting with a VAR

A VAR forecast pushes all variables one step ahead at once with a single matrix multiplication, then feeds that forecast back in and repeats.

One-step-ahead forecast

Simply plug in the latest observed values of all variables. The error has expectation 0, so it drops out.

$$\hat{\mathbf{Y}}_{t+1|t} = \mathbf{c} + A_1\mathbf{Y}_t + A_2\mathbf{Y}_{t-1} + \cdots + A_p\mathbf{Y}_{t-p+1}$$

Multi-step forecasts: iterate

For h > 1, put the previous step's forecast in place of values not yet observed. For a VAR(1) it reads as follows.

$$\hat{\mathbf{Y}}_{t+h|t} = \mathbf{c} + A\,\hat{\mathbf{Y}}_{t+h-1|t}$$

For a stable VAR, the forecast converges to the long-run mean as h grows. The long-run mean is found by solving μ = c + Aμ.

$$\mu = (I - A)^{-1}c$$

Forecast intervals widen together

The covariance of the h-step forecast error is the sum of the shocks accumulated in the meantime.

$$\Sigma(h) = \sum_{i=0}^{h-1} A^i \,\Sigma\, \left(A^i\right)^\top$$

Because a term is added at each step, the interval widens as the horizon lengthens. If the VAR is stable, A^i shrinks to 0 and the interval levels off at a fixed width. Since the off-diagonal elements of Σ enter this formula, the intervals of the two variables move **together (jointly)**.

Why joint forecasting is better

- Each variable's forecast makes use of the other variables' recent movements.
- The forecasts are mutually consistent and respect the estimated cross-dynamics.
- All variables are forecast at once.

8. The Five-Step VAR Workflow

There are five steps from data to a fitted system: steps 1–4 build the model, and step 5 is the analysis of Parts 2 and 3.

1. **Make stationary**: run a unit-root test on each series and, if needed, difference it into a stationary series.
2. **Choose the lag**: compare AIC, BIC and HQ to pick p. Keep p small.
3. **Estimate**: obtain A and the residual covariance Σ by equation-by-equation OLS.
4. **Diagnose**: check that all eigenvalues lie inside the unit circle (stability) and that the residuals have no autocorrelation.
5. **Use**: produce joint forecasts and analyse with Granger causality, IRF and FEVD.

Univariate models compared with a VAR

Aspect	Separate univariate models	VAR (joint)
Cross-dynamics	Ignored	Captured by off-diagonal coefficients
Forecasts	Separately for each variable	All variables together, consistently
Lead–lag and feedback	Invisible	Explicit and testable
Parameter burden	Light	Heavy (proportional to k^2)

A VAR captures interactions realistically but uses far more parameters. That is why k and p are kept small.

9. Common Misconceptions and Self-Check

Common misconception	Reality
The more variables you add, the better the VAR	Each extra variable makes the parameters grow as k^2. Keep the VAR small (Section 6)
A VAR shows what causes what	A reduced-form VAR shows only predictive

Common misconception	Reality
	relationships; it does not prove causation (Section 4, Part 2)
You can fit a VAR directly to price levels	A standard VAR needs stationary series. Difference first, or use a VECM if the series are cointegrated (Section 5, Week 7)

Three questions before moving on

1. **In a two-variable VAR, what does a non-zero off-diagonal coefficient mean?** It means the past of one variable helps forecast the other.
2. **How do you check whether a fitted VAR is stable?** Check that all eigenvalues of the coefficient matrix (the companion matrix, if there are several lags) lie inside the unit circle.
3. **Why must the number of variables in a VAR be kept small?** Because the number of parameters, $k + pk^2$, grows with the square of the number of variables.

Next: Part 2

With Granger causality, we test whether the past of one variable **really** improves the forecast of another. The core question is whether the off-diagonal coefficients a12 and a21 from Section 2 are statistically zero.